

ShiftGuard10: Robust Image Classification

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Abstract—In this Project, an algorithm that utilizes deep learning to perform multi-class image recognition by means of Residual Neural Networks (ResNet) is introduced. This algorithm is intended to recognize distorted and misaligned images. Data augmentation and test-time augmentation are used in the proposed model to make it more efficient than the existing CNN-based methods.

I. INTRODUCTION

Robust image classification is the problem posed for our project. The images have to be classified even when they are subjected to shift, distortion, or noise. This is obviously difficult compared to image classification tasks since there are a number of distortions that need to be accounted for. In order to address this problem, CNN is used. This neural network architecture is useful for identifying image features such as edges and textures through learning. Moreover, for enhanced efficiency, Residual Neural Network (ResNet) is designed using skip connections. In this way, the depth of learning is facilitated. Furthermore, data augmentation techniques have been used in order to increase the efficiency of the learning process by making it robust to changes.

II. DATASET AND PREPROCESSING

Training data are comprised of training images with labels and testing images without labels, both of which perform multiclass classification. The images are transformed into RGB format and normalized to enhance the training process. As depicted in Fig. 1, the dataset is slightly unbalanced, meaning that the distribution of some classes is higher than other classes. Therefore, stratified sampling was applied to split the dataset into training and validation datasets. Labels were transformed into one-hot encoding.

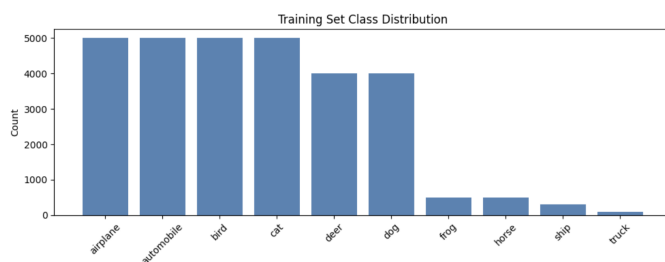


Fig. 1. Training set class distribution.

III. DATA AUGMENTATION

For better generalization, some data augmentation methods such as flipping, rotating, zooming, and shifting are used. Through these processes, the model learns from various kinds of variations of data.

IV. MODEL ARCHITECTURE

The network follows the structure of ResNet. This model contains several residual blocks where the convolution, batch normalization, and ReLU activation are utilized. The skip connection makes it easier to transmit gradients and enables deep learning. In the proposed architecture, the filters are 64, 128, 256, and 512. The global average pooling is used before reaching the fully connected layer. For regularization, the dropout technique is employed. The final layer uses the softmax function.

V. TRAINING STRATEGY

Training of the model is done through Stochastic Gradient Descent (SGD) with momentum and Nesterov acceleration. Cosine learning rate annealing is employed to lower the learning rate during training. Weight decay regularization technique is employed to mitigate overfitting of the model. Early stopping and saving of checkpoints are done as means of saving the best model. A batch size of 64 is chosen for training the model.

VI. RESULTS AND ANALYSIS

The performance of the trained model is measured through accuracy, loss, and F1 score. The training and validation curves are illustrated in Fig. 2.



Fig. 2. Accuracy and loss curves during training.

Accuracy rises gradually, reflecting learning. The training accuracy approaches 1.0, whereas validation accuracy converges to a little smaller value, reflecting successful generalization. The loss declines continually, where the gap between training and validation losses is minor, implying a little overfitting.

A. Confusion Matrix

The confusion matrix is shown in Fig. 3. Most predictions lie along the diagonal, indicating correct classifications. Some errors occur between similar classes.

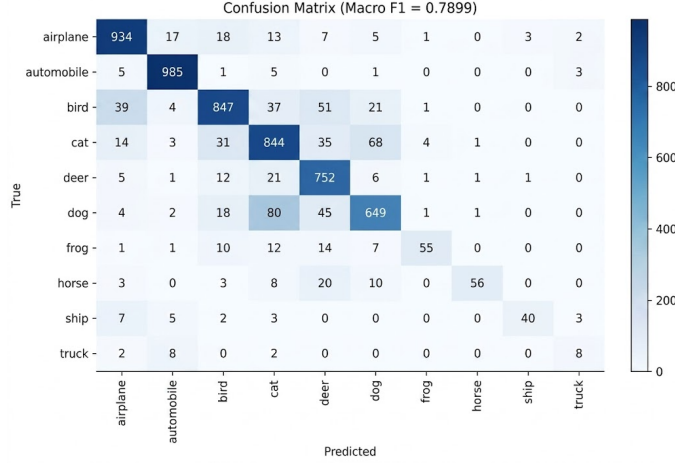


Fig. 3. Confusion matrix of validation predictions.

B. Classification Metrics

The F1-score for each label is shown in Fig. 4. The model performs consistently across most classes with only slight variations, except truck and frog because we have very few samples of them.

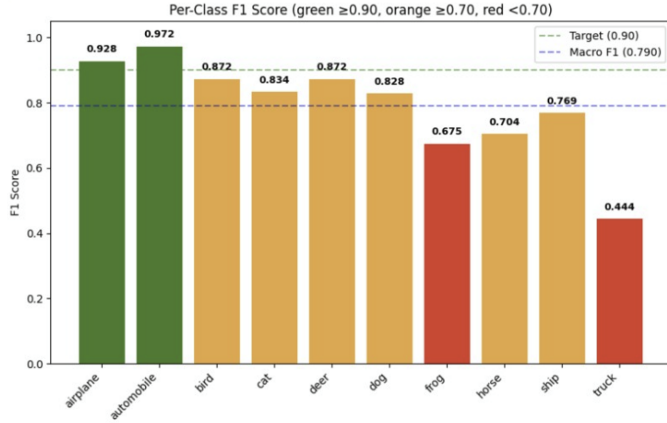


Fig. 4. F1-score of each label

C. Test-Time Augmentation

Test-time augmentation (TTA) is applied by generating multiple augmented versions of each test image and averaging predictions. This improves stability and overall accuracy.

VII. CONCLUSION

ResNet architecture has been used in this project for effective image classification. Data Augmentation and Regularization improved generalization, while test time augmentation improved prediction stability. The performance of the model is

optimal with a low level of overfitting and consistent accuracy in all classes. Further improvements can be made by using ensembles of models and other advanced architectures..

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